

# Confidence Intervals and the Z-procedure

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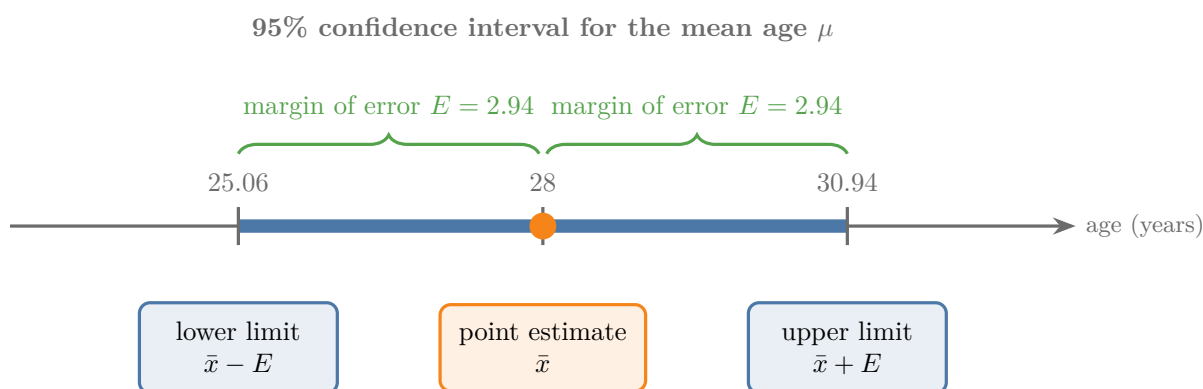
Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Standard normal and the z-table ([Note 251](#)); Central limit theorem ([Note 272](#)).
- **Compare with:** Hypothesis testing: null and alternative ([Note 290](#)).

## 1. Overview

**Key point:** A confidence interval replaces a single guess with a range: point estimate  $\pm$  margin of error, together with a confidence level such as 95%.



**Figure 1:** A 95% confidence interval for a mean age: the point estimate in the middle, the margin of error on each side

A sample mean is almost never exactly the population mean. A **confidence interval** admits this: it gives a range of plausible values for the parameter and says how confident we are in the method that built it. Figure 1 shows its parts for one example worked out in this Note.

This Note covers:

- why a point estimate alone is not reliable;
- what a confidence interval and a confidence level are;
- the **z-procedure**, used when the population standard deviation  $\sigma$  is known: its assumptions, its formula, where the formula comes from, and how to find the critical value  $z_{\alpha/2}$ .

The [interpreting confidence intervals Note](#) explains what “95% confident” really means and what makes an interval wide or narrow. The [t-procedure Note](#) handles the usual case where  $\sigma$  is unknown.

## 2. Terms we build on

**Key point:** Population and sample, parameter and statistic, inferential statistics, the central limit theorem and the point estimate.

Each of these is taught in an earlier Note; here is the one-line version.

- **Population and sample:** the population is the entire group we want to study; a sample is the part we measure, chosen at random and representative (see the [what is statistics Note](#)).
- **Parameter and statistic:** a number describing the population (Greek letters:  $\mu, \sigma$ ) and the same number computed from a sample (Latin letters:  $\bar{x}, s$ ). Parameters are usually unknown and must be estimated from statistics.
- **Inferential statistics:** drawing conclusions about a population from a sample. Its tools include confidence intervals, hypothesis tests and regression, and it answers questions such as “can the sample mean tell us the population mean?”.
- **Central limit theorem:** the means of large samples are approximately normal around  $\mu$  with standard error  $\sigma/\sqrt{n}$  (see the [central limit theorem Note](#)).
- **Point estimate:** a single number computed from sample data as the best guess for an unknown parameter (see the [estimating a mean Note](#)).

The running example of this Note: an online channel has about 77,000 subscribers, and we want their average age  $\mu$ . Asking all of them is impossible. Instead, 100 subscribers in an online event type their age in the chat. Their mean age is  $\bar{x} = 28$  years: a point estimate of  $\mu$ .

Averaging the means of 10 such events would give a better point estimate, as in the estimating a mean Note. The average would still be a single number.

## 3. Why a point estimate is not enough

**Key point:** An exact guess is almost always wrong; a range is far more likely to be right.

Suppose we must guess how many runs MS Dhoni will score in today’s match, with three ways to bet:

| Bet                  | Wins if                             | Prize          |
|----------------------|-------------------------------------|----------------|
| Exact score          | the guess is exactly right          | 1 crore rupees |
| Within $\pm 10$ runs | the score is within 10 of the guess | 75 lakh rupees |
| Within $\pm 20$ runs | the score is within 20 of the guess | 50 lakh rupees |

A sensible person takes the second or third bet. Hitting the exact score is very unlikely, while a range of 20 or 40 runs has a real chance. If we guess 35 and he scores 28, the difference is 7: the  $\pm 10$  bet wins.

The same holds for estimates. Claiming that the mean age of 77,000 people is exactly 28, because 100 of them average 28, is not believable. So statisticians report a range that probably contains the parameter.

## 4. Confidence intervals and confidence levels

**Key point:** A confidence interval is a range of plausible values for a population parameter; the confidence level, usually 95%, says how often the method that builds such intervals succeeds.

A **confidence interval (CI)** is a range of values within which we expect a population parameter, such as  $\mu$  or  $\sigma$ , to lie. The interval expresses the uncertainty of an estimate obtained from a sample.

Every confidence interval comes with a **confidence level**, a percentage such as 95%. “The mean age of the subscribers is between 25 and 32 years, at 95% confidence”: the range 25 to 32 is the interval, and 95% is the level. The exact meaning of the level is subtle; the [interpreting confidence intervals Note](#) is devoted to it.

Every confidence interval has the same structure.

1. **In words:** start from the point estimate and go one **margin of error** down and one up.
2. **Formula:**

$$\text{CI} = \text{point estimate} \pm \text{margin of error}$$

3. **Example:** a point estimate of 25 years and a margin of error of 4 years give

$$25 \pm 4 : \quad 25 - 4 = 21 \text{ (lower limit)}, \quad 25 + 4 = 29 \text{ (upper limit)}$$

So building a confidence interval needs two things: a point estimate and a margin of error. The rest of this Note is about the margin of error.

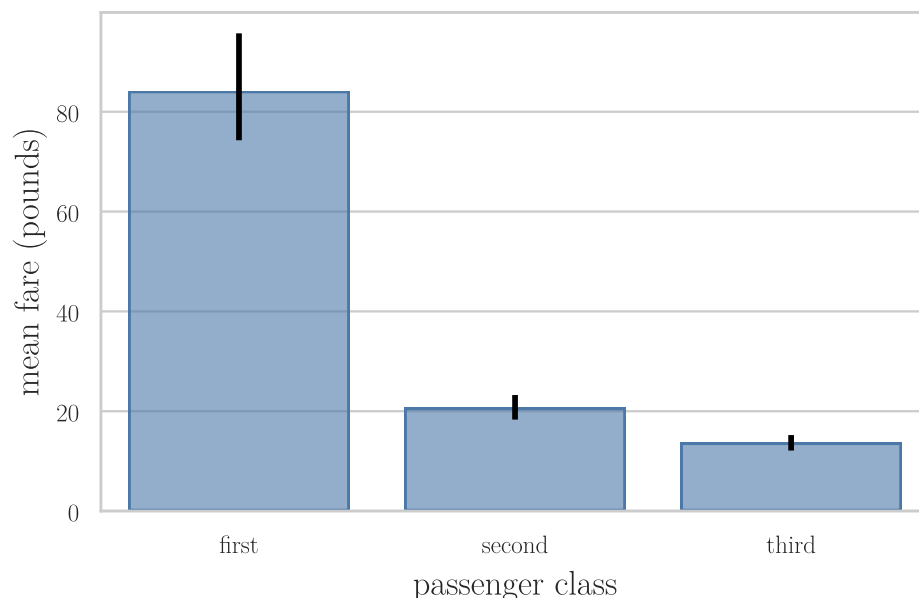
A confidence interval is always about a **parameter** of the population (here  $\mu$ ), never about a statistic. The statistic  $\bar{x}$  is known exactly; it is the tool we use to build the interval.

### 4.1 Confidence intervals in practice

**Key point:** Business forecasts and the error bars that seaborn draws on a bar chart are confidence intervals.

Companies, finance and economics rarely report a single projection. A sales forecast is a range: “between 4.2 and 4.8 crore rupees next quarter”.

We have also drawn confidence intervals already without naming them. Figure 2 plots the mean Titanic fare of each passenger class with seaborn, as in the [bivariate analysis Note](#). The black line on top of each bar is a 95% confidence interval for the mean fare of that class.



**Figure 2:** Mean Titanic fare per class (bars) with 95% confidence intervals (black lines)

The bars show the sample means; the lines show where the population means of passengers like these probably lie. First class has 216 passengers whose fares are very spread out (standard deviation about 78 pounds), and its interval is wide (about 74 to 95 pounds). Third class has 491 passengers with similar fares (standard deviation about 12 pounds), and its interval is narrow (about 12.6 to 14.7 pounds). Both match the standard error  $s/\sqrt{n}$  of section 7: a larger spread or a smaller sample gives a wider interval.

**Extra:** Seaborn builds these intervals by bootstrapping (see the [bagging Note](#)): it resamples the data with replacement (`n_boot=1000` times by default), recomputes the mean each time, and keeps the middle 95% of those means (Waskom 2021; seaborn docs v0.13). In the notebook, the formula of this Note with  $s$  in place of  $\sigma$  gives almost the same ranges: 73.7 to 94.6 pounds for first class (seaborn: 74.3 to 95.2) and 12.6 to 14.7 pounds for third class (seaborn: 12.7 to 14.8).

## 5. Two ways to compute a confidence interval for a mean

**Key point:** The z-procedure needs the population standard deviation  $\sigma$ ; the t-procedure works with the sample standard deviation  $s$ .

There are several ways to compute a confidence interval. For a mean, the two standard ones are:

| Procedure                          | Use when                                            | Spread used |
|------------------------------------|-----------------------------------------------------|-------------|
| <b>Z-procedure</b> (“sigma known”) | the population standard deviation $\sigma$ is known | $\sigma$    |

| Procedure                            | Use when            | Spread used                       |
|--------------------------------------|---------------------|-----------------------------------|
| <b>T-procedure</b> (“sigma unknown”) | $\sigma$ is unknown | the sample standard deviation $s$ |

In practice  $\sigma$  is almost never known: if we do not know the mean age of 77,000 subscribers, we hardly know the standard deviation of their ages. So the t-procedure is the one used in real work. We start with the z-procedure; the t-procedure has the same form, with  $s$  and a t critical value in place of  $\sigma$  and  $z$  (see the [t-procedure Note](#)).

## 6. Assumptions of the z-procedure

**Key point:** A random sample, a known  $\sigma$ , and a normal population or a sample large enough for the central limit theorem.

The z-procedure gives correct intervals only when three assumptions hold:

1. **Random sample.** The sample is drawn at random and is representative. A sample of only Indian subscribers says little about a channel with subscribers in many countries.
2. **Known population standard deviation.** We know  $\sigma$ , here the standard deviation of the ages of all subscribers. We assume  $\sigma = 15$  years.
3. **Normal sampling distribution.** Either the population itself is normal, or the sample is large, usually  $n > 30$ . Then, by the central limit theorem, the sample mean is approximately normal whatever the shape of the population (see the [central limit theorem Note](#)).

The third assumption is therefore mild. A population of ages with three peaks, at 26, 32 and 44, is clearly not normal; with a sample of 100 the z-procedure still works. With a non-normal population and only 20 people in the sample, it does not: neither the population nor the CLT gives a normal sample mean.

## 7. The z-procedure formula

**Key point:**  $\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$ , where  $1 - \alpha$  is the confidence level and  $z_{\alpha/2}$  is read from the standard normal curve.

The problem: a sample of  $n = 100$  subscribers has mean age  $\bar{x} = 28$  years, the population standard deviation is  $\sigma = 15$  years, and we want a 95% confidence interval for  $\mu$ .

1. **In words:** go from the sample mean  $z_{\alpha/2}$  standard errors down and up. The standard error is  $\sigma / \sqrt{n}$ , and the margin of error is  $z_{\alpha/2}$  times it.
2. **Formula:**

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$ : the sample mean, our point estimate.
- $1 - \alpha$ : the confidence level. For 95%,  $1 - \alpha = 0.95$ , so  $\alpha = 0.05$  and  $\alpha/2 = 0.025$ .

- $z_{\alpha/2}$ : the **critical value**, the z-score that leaves an area of  $\alpha/2$  in the upper tail of the standard normal curve. For 95% it is 1.96 (section 9).
- $\sigma$ : the population standard deviation;  $n$ : the sample size.

### 3. Example:

$$SE = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5, \quad E = 1.96 \times 1.5 = 2.94$$

$$28 \pm 2.94 : \quad 28 - 2.94 = 25.06, \quad 28 + 2.94 = 30.94$$

The 95% confidence interval for the mean age of all subscribers is **25.06 to 30.94 years** (Figure 1).

The formula leaves two questions: why this formula, and how to find  $z_{\alpha/2}$ . The next two sections answer them.

**Python:** The z-interval with scipy.

```
import numpy as np
from scipy import stats

x_bar, sigma, n = 28, 15, 100
se = sigma / np.sqrt(n)                # 1.5
z = stats.norm.ppf(0.975)              # 1.96
x_bar - z * se, x_bar + z * se          # (25.06, 30.94)
stats.norm.interval(0.95, loc=x_bar, scale=se) # same
```

## 8. Where the formula comes from

**Key point:** Standardize the sample mean, find the middle 95% of the standard normal curve, and rearrange the inequality so that  $\mu$  stands alone.

**Step 1: the sample mean is normal.** By the assumptions (or the CLT),  $\bar{X}$  follows a normal distribution with mean  $\mu$  and standard deviation  $\sigma/\sqrt{n}$ . Imagine many online events of 100 subscribers each: their mean ages  $\bar{x}_1, \bar{x}_2, \dots$  form this bell (see the [central limit theorem Note](#)).

**Step 2: standardize.** Subtracting the mean and dividing by the standard deviation turns any normal variable into the standard normal  $Z \sim N(0, 1)$  (see the [standard normal Note](#)):

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

**Step 3: the middle  $1 - \alpha$  of  $Z$ .** We want a range of  $z$  values that contains  $Z$  with probability  $1 - \alpha$  (95%). The curve is symmetric, so the remaining  $\alpha$  splits into  $\alpha/2$  in each tail (Figure 3). The two cut-off points are  $-z_{\alpha/2}$  and  $+z_{\alpha/2}$ :

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

**Step 4: put  $\bar{X}$  back in and isolate  $\mu$ .**

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

Multiply all three parts by  $\sigma/\sqrt{n}$ :

$$P\left(-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Subtract  $\bar{X}$  from all three parts:

$$P\left(-\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < -\mu < -\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Multiply by  $-1$ , which flips both inequality signs:

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

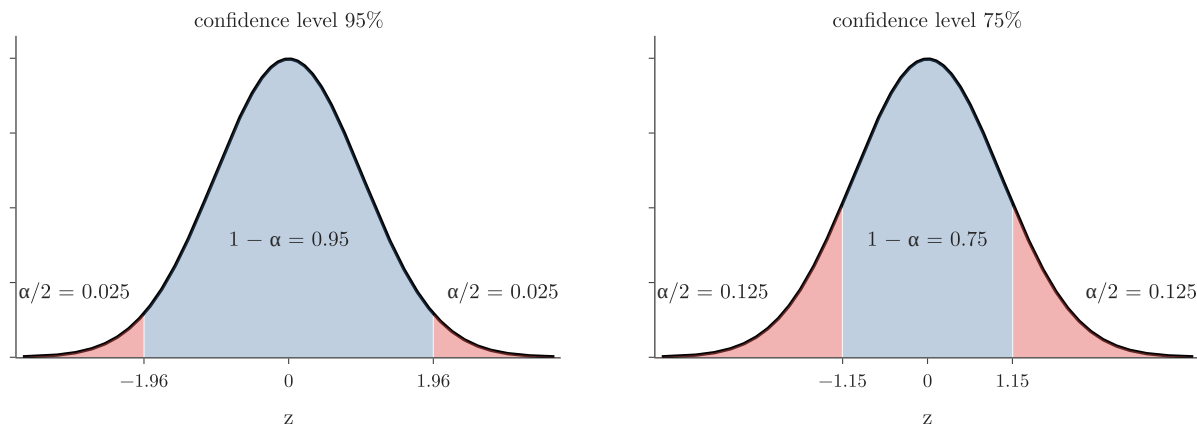
The two ends are exactly the confidence interval  $\bar{X} \pm z_{\alpha/2} \sigma/\sqrt{n}$ .

**Step 5: what the probability belongs to.** The last line looks like “ $\mu$  lies in this range with probability 95%”, but  $\mu$  is a fixed number: the true mean age of the subscribers does not vary. What varies is  $\bar{X}$ : the next online event brings different people and a different mean. The probability belongs to the random interval, not to  $\mu$ .

So once we compute one interval from one sample (25.06 to 30.94), we do not say “ $\mu$  is in it with probability 95%”. We say we are **95% confident**: 95% of intervals built this way contain  $\mu$ . The [interpreting confidence intervals Note](#) shows this with a simulation.

## 9. Finding the critical value $z_{\alpha/2}$

**Key point:** Look up the area  $1 - \alpha/2$  in the z-table: 0.975 gives  $z = 1.96$  for 95%, and 0.875 gives  $z = 1.15$  for 75%.



**Figure 3:** The middle  $1 - \alpha$  of the standard normal curve and the critical values:  $\pm 1.96$  for 95%,  $\pm 1.15$  for 75%

A z-table gives the area to the **left** of  $z$  (see the [standard normal Note](#)). The area to the left of  $+z_{\alpha/2}$  is the middle  $1 - \alpha$  plus the left tail  $\alpha/2$ .

1. **In words:** add the left tail to the middle area, then find the  $z$  with that area to its left.
2. **Formula:**

$$\Phi(z_{\alpha/2}) = (1 - \alpha) + \frac{\alpha}{2} = 1 - \frac{\alpha}{2}$$

3. **Example:** for 95%,  $\alpha = 0.05$ :

$$\Phi(z_{0.025}) = 0.95 + 0.025 = 0.975$$

The z-table has 0.9750 in row 1.9, column 0.06, so  $z_{0.025} = 1.96$ . By symmetry the lower cut-off is  $-1.96$ .

For a 75% confidence level,  $\alpha = 0.25$  and each tail holds 0.125. The area to the left is  $0.75 + 0.125 = 0.875$ , which the z-table places at  $z = 1.15$  (Figure 3, right). The 75% interval for the subscribers is  $28 \pm 1.15 \times 1.5 = 28 \pm 1.73$ : 26.27 to 29.73 years, narrower than the 95% interval.

| Confidence level | $\alpha/2$ | Area to the left | $z_{\alpha/2}$ |
|------------------|------------|------------------|----------------|
| 50%              | 0.25       | 0.75             | 0.674          |
| 75%              | 0.125      | 0.875            | 1.150          |
| 80%              | 0.10       | 0.90             | 1.282          |
| 90%              | 0.05       | 0.95             | 1.645          |
| 95%              | 0.025      | 0.975            | 1.960          |
| 99%              | 0.005      | 0.995            | 2.576          |

The critical value completes the z-procedure: with  $\bar{x}$ ,  $\sigma$ ,  $n$  and the critical value for the chosen confidence level, the interval follows from one formula. 95% is the usual level; the next Note shows why.

**Python:** Critical values with scipy.

```
from scipy import stats
```

```
level = 0.95
```

```
stats.norm.ppf(1 - (1 - level) / 2)    # 1.96
```

```
stats.norm.ppf(0.875)                  # 1.15, for 75%
```

ppf is the inverse of the CDF: it takes an area and returns the  $z$  with that area to its left.

## 10. Summary

| Idea                | Formula                                      | Example                  |
|---------------------|----------------------------------------------|--------------------------|
| Confidence interval | point estimate $\pm$ margin of error         | $25 \pm 4$ : 21 to 29    |
| Confidence level    | $1 - \alpha$                                 | 0.95, so $\alpha = 0.05$ |
| Standard error      | $\sigma / \sqrt{n}$                          | $15 / \sqrt{100} = 1.5$  |
| Critical value      | $\Phi(z_{\alpha/2}) = 1 - \alpha/2$          | $\Phi(1.96) = 0.975$     |
| Margin of error     | $E = z_{\alpha/2} \sigma / \sqrt{n}$         | $1.96 \times 1.5 = 2.94$ |
| Z-interval          | $\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$ | 25.06 to 30.94 years     |

- A point estimate is almost never exactly right; an interval states the uncertainty.
- A confidence interval is about a population parameter, built from sample statistics.
- The z-procedure needs a random sample, a known  $\sigma$ , and a normal population or  $n > 30$ .
- The probability of 95% belongs to the method (the random interval), not to the fixed  $\mu$ .

## 11. Sources

### Built from

- CampusX, “Session 44 - Confidence Intervals | DSMP 2023”, YouTube, <https://www.youtube.com/watch?v=...>

### Other references

- Waskom, M. L. (2021). “seaborn: statistical data visualization.” *Journal of Open Source Software* 6(60), 3021.
- seaborn developers. *seaborn v0.13 documentation*: tutorial “Statistical estimation and error bars”, and API reference `seaborn.barplot (errorbar=("ci", 95), n_boot=1000)`. [seaborn.pydata.org](https://seaborn.pydata.org).

## 12. Key terms

| Term                  | Meaning                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Confidence interval   | A range of plausible values for a population parameter, computed from a sample: point estimate $\pm$ margin of error |
| Confidence level      | The share of intervals built by the method that contain the parameter, such as 95%; written $1 - \alpha$             |
| Margin of error       | The distance from the point estimate to either end of a confidence interval                                          |
| Lower and upper limit | The two ends of a confidence interval                                                                                |
| Z-procedure           | The confidence interval $\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$ , used when $\sigma$ is known                   |

| Term           | Meaning                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------|
| Critical value | The z (or t) value that leaves $\alpha/2$ in each tail; 1.96 for 95% on the standard normal curve |
| $\alpha$       | One minus the confidence level: the share of intervals that miss the parameter                    |